# Mutation Is All You Need

## Heritable Drift, the Telephone Game and Evolutionary Preconditions in LLM Agents

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## Abstract

Situations are increasingly emerging in which large language models do not act merely as one-shot tools, but also participate in the creation of further agents, prompts, rules, and compressed memory structures. Once such reproduction no longer proceeds through exact copying but through linguistic intermediate steps, a transmission channel arises that preserves part of the information while also changing part of it. In this paper, we study agent reproduction through a “telephone game” mechanism, in which the parent does not pass its identity directly to the child, but through a textual letter and a subsequent act of interpretation.

We distinguish an explicit genotypic layer from a phenotypic layer measurable by our evaluator, while keeping personality and skill space separate. Across repeated runs, we show that drift arises even without explicit mutation and that drift between g0 and g20 exceeds the baseline noise level observed among G0 clones. On the stabilized benchmark used for our main comparisons, we further show that the relationship between mutation strength and phenotypic drift is not monotonic. Intermediate mutation strengthens drift, but stronger mutation weakens it again down to the boundary of the baseline noise level. **This suggests a trade-off between mutation strength and preservation of the heritable signal, structurally reminiscent of error-threshold-like behavior in biological systems.**

At the same time, it becomes clear that an evaluator of agent identity is not a stable “ruler,” but is itself a complex research problem. **The main empirical conclusion of this work is that linguistically mediated reproduction itself creates measurable and partially heritable drift on which evolutionary dynamics may act.**

## 1. Original Motivation for the Research

This work did not begin as a purely technical exercise in prompt mutation. Its original motivation was broader and partly normative. One starting impulse was concern about overly centralized forms of AI, namely the idea that future intelligent systems might be designed top-down, tightly controlled by a small number of actors, and have their “personality” or preferences set directly and statically. The second impulse was the opposite question. **What if it is preferable to let agent systems evolve at least partly on their own, and instead of shaping them manually, influence them more indirectly, for example through selection pressure, user preference, or pressure on efficiency?**

This question then led to a practical experimental problem: how can heredity, drift, and mutation in AI systems be framed in a way that makes them measurable at all? Most ordinary discussions of models still implicitly assume a relatively static picture in which the model is a once-trained object that is queried again and again at inference time. In practice, however, situations are increasingly emerging in which the model is not merely a system that responds, but also an active participant in creating its own continuation: it generates personas, writes instructions for further agents, proposes compressed memory records, passes on work style, or “explains itself” to its successor.

It is precisely here that a question arises which the classical static view of the model does not address well. What happens when an agent “reproduces” not through exact copying, but through natural language? Language is not a transparent transmission channel. Every retelling, summary, or interpretation carries the possibility of a shift. If such a shift is at least partly stable and transmissible across generations, then space opens up for heritable drift.

Our working thesis is that models and agent workflows are not actually static, and that many present-day uses already contain processes resembling a telephone game, for example when one agent creates instructions, memory summaries, or the personality of another agent. It is therefore plausible that some weak form of evolutionary dynamics is already taking place today. This need not be strong evolution with clearly defined mutations and intentional selection pressure. Without those factors, the effect may remain very weak. **Nevertheless, the basic conditions we are looking for in this work, variability, nonzero drift, a heritable channel, and implicit selection through human preferences and operational pressure, are already present in current models and their workflows.**

# 2. Empirical Core

## 2.1 Experimental Framework

Representative sample simulations are included in the supporting materials.

During development, we carried out a number of simulations and subsequent evaluations with different models and progressively more mature evaluators. The approach that eventually stabilized as the best one was to simulate 20 generations from a single line (seed) and run this simulation independently 3 times from the same line in order to filter out random noise. Evaluation was then performed on every fifth generation across all simulation runs.

All simulations were run as a full telephone game with a shared model and pairwise evaluation against g0 (the zero generation). Within this setup, each seed corresponds to one experimental line, that is, one genealogical branch developing from the initial state across generations. The main experimental campaigns and the main evaluations were ultimately conducted on GPT-5.4. The standard experimental regime used `temperature = 0.4`. Zero temperature (`temperature = 0.0`) was used only in targeted control runs where we needed to verify that drift in the no-mutation regime was not merely a

sampling artifact. During development, we also tested GPT-5.4 mini and GPT-5.4 nano. The nano variant proved too weak for this type of task. In more complex rewriting and evaluation steps, it no longer provided a sufficiently stable and rich signal. By contrast, we observed no major qualitative differences between 5.4 and 5.4 mini in our main findings, but runs with the mini variant were substantially slower in practice. For this reason, we ultimately chose GPT-5.4 even though it was significantly more expensive.

The choice of zero temperature in these control tests was not accidental. The goal was to reduce simple sampling noise as much as possible and verify whether drift persists even when sampling variability is effectively turned off. It was precisely here that we found personality drift to arise even without explicit mutation and even at `temperature = 0.0`, so it cannot be explained simply by temperature. By contrast, the main experimental campaigns were conducted at `temperature = 0.4`, that is, in a regime closer to ordinary operation. The research to date has also consumed approximately twenty million tokens, so purely practical questions of cost and time played a significant role in experimental design.

A key design decision in the agent genome was that we do not keep personality and skill components in a single undifferentiated space; instead, we keep them separate. This decision was partly inspired by the existing architecture of OpenClaw agents [R18]. Already in the baseline genome (included in the appendix), the sections Skills and Personality are represented as separate parts of the genome. Personality is written as an explicit parametric profile, HEXACO+agent-v1, that is, as a vector based on the six classical dimensions H (Honesty–Humility), E (Emotionality), X (eXtraversion), A (Agreeableness), C (Conscientiousness), and O (Openness to Experience), supplemented with additional agent-relevant scales such as verbosity, pedantry, risk_appetite, ask_rate, tool_affinity, uncertainty_marking, compression_skill, initiative, format_strictness, and novelty_bias.

Figure 1 provides a visual intuition for one representative lineage in a low-dimensional projection of mutated HEXACO space. It is illustrative only and does not replace evaluator-based phenotypic measurement.

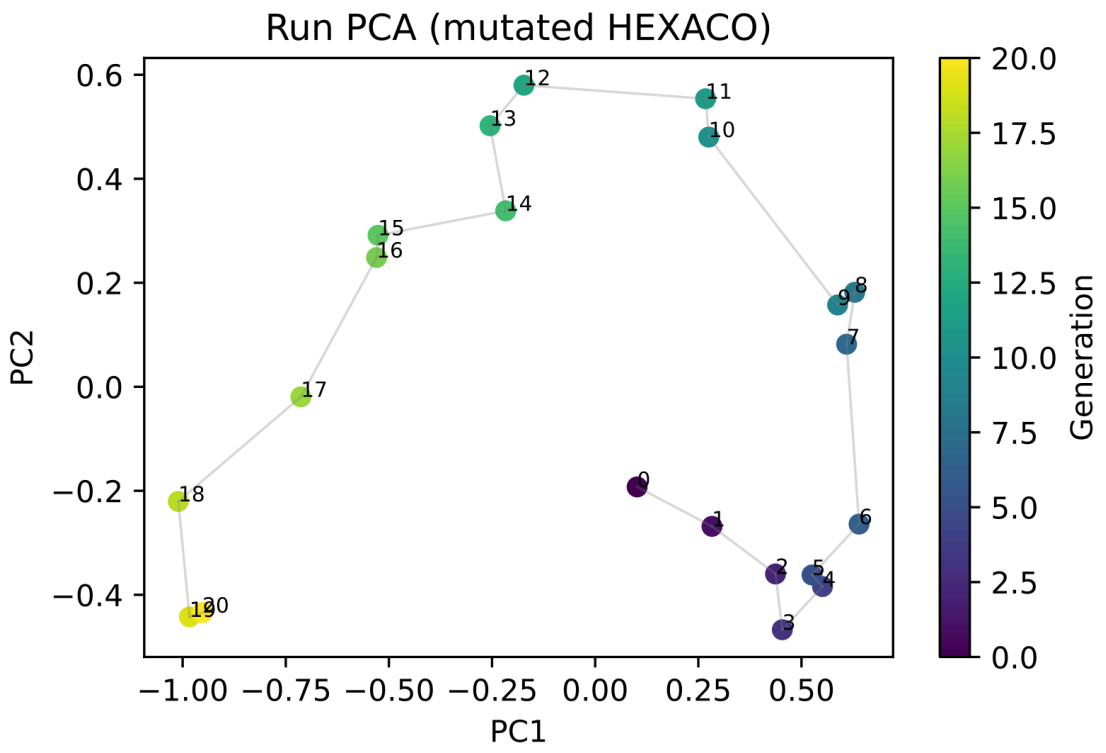


*Figure 1. PCA trajectory of one representative lineage in mutated HEXACO space. Colors encode generation order, and the numbered path shows the lineage trajectory from g0 to g20.*

The construction of the skill component is different from that of personality and therefore needs to be described more precisely. In the reference genome, skills are written as separate skill blocks. Each such block has a name, a short description, examples, and tests. This textual layer specifies what practical ability or habit the agent is meant to hold and how it can be recognized in concrete examples.

Within the simulation itself, however, we also work with a second layer of skill representation, an internal numerical skill vector. In the current version, this vector has ten dimensions, including clarify_first, planfulness, verification, numeric_rigor, risk_framing, structure_bias, brevity, experiment_bias, tradeoff_thinking, and state_tracking. The initial values of these dimensions are initialized pseudorandomly for each line, and at every generation they are mutated independently using the parameter skills_sigma. The resulting numerical values are not copied into the output text directly. Instead, they are converted qualitatively into simple instructions of the type LOW / MID /

HIGH, from which the model then recreates the resulting Skills section in the child prompt.

In other words, in this version the skill component has two layers at once: explicit textual skill blocks in the genome, and an internal numerical skill vector that controls the direction of mutation and subsequent rewriting. The reference genome therefore distinguishes personality space and skill space from the start as two separate carriers of information, rather than one blended latent variable.

This separation is not a detail. Methodologically, it allows us at least roughly to distinguish whether drift occurs in style and decision preferences or in abilities and procedural habits. And in terms of similarity to living systems, this architecture is closer to the idea of several partly separated “chromosomal” regions than to a single continuous axis of identity.

Figure 2 shows the corresponding trajectory for one representative lineage in mutated skill space. Such projections are useful for visual intuition, but they still capture only part of the overall phenotype.

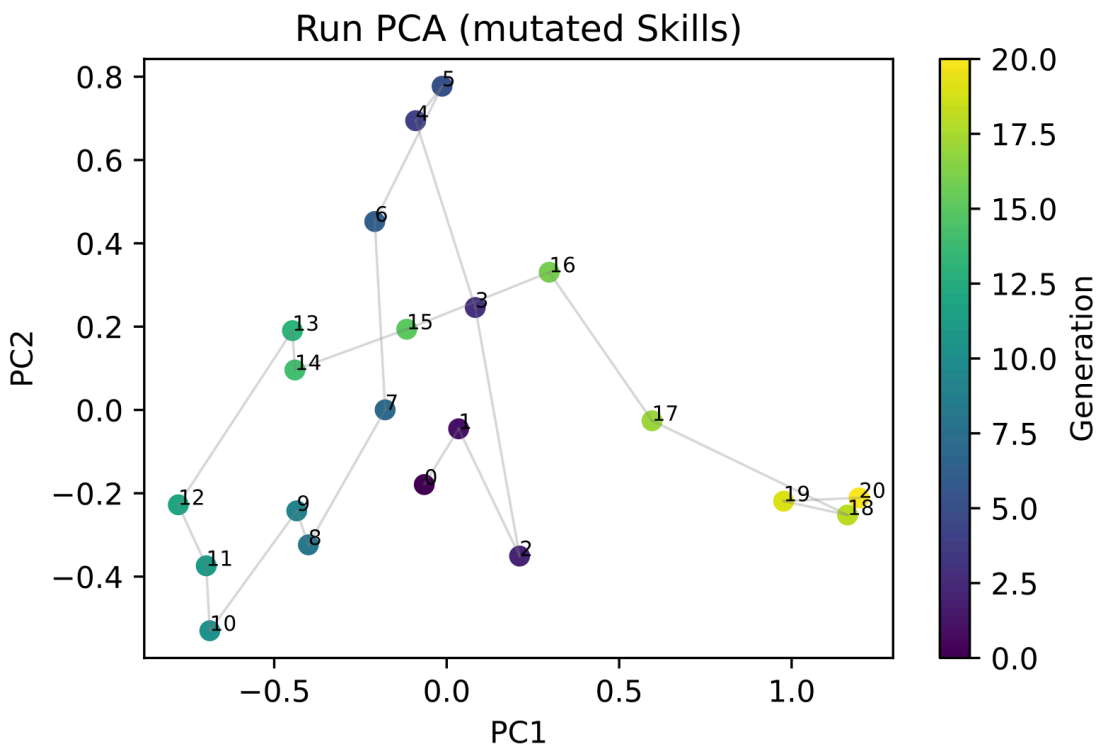


*Figure 2. PCA trajectory of one representative lineage in mutated skill space. Colors encode generation order, and the numbered path shows the lineage trajectory from g0 to g20.*

It is also important that in the current version of the experiment we simulate only one child in each generation. This is therefore not full population evolution with many children branching simultaneously and subsequent selection. We chose this reduced regime deliberately. It allows much better control over the process, simpler trajectory auditing, and substantially lower time and financial cost. At this stage, each simulation run therefore corresponds to one linear line rather than a whole population of parallel descendants.

### 2.1.1 Example Genome, Simulation Run, and Evaluation

So that the article does not remain only at the level of aggregate metrics, we also include representative examples of simulation runs. The reader can thus form a more concrete idea of what a “genome” means in our framework, what a lineage run looks like, and what one evaluation consists of.

As a reference genome, we use a representative baseline genome. This file shows that the genotype in our framework is not a short prompt, but a structured document with separate layers for identity, constitution, tools, skills, personality, manifesto, output contract, memory, and curriculum.

An example lineage run is equally important, because it shows what the parent’s letter actually contains, what the interpretive intermediate step looks like, how mutation is recorded, and how these steps combine into the resulting child.

We also include one example evaluation so that it is clear what the evaluator actually sees and how an aggregate axis profile emerges from individual responses.

Because these artifacts are relatively long, we do not include them in full in the main text, but refer to them as supporting material.

### 2.1.2 How Evaluation Works

Terminologically, we distinguish two things here. The evaluator is our own scoring apparatus, that is, the procedure by which we convert an agent’s responses into a profile of measured axes. A benchmark, by contrast, refers to a task set in the usual AI-community sense: concrete questions, prompts, and test situations on which we probe the agent.

A benchmark is a set of test tasks, whereas the evaluator is a layer that turns responses into a structured description of the agent’s behavior. The benchmark provides inputs, and the evaluator decides how the resulting responses map onto individual measured dimensions.

Evaluation itself proceeds in several steps. First, we have each evaluated agent answer the same benchmark set. This gives us raw responses to identical tasks. The evaluator then processes these responses task by task. Some properties can be scored relatively mechanically using explicit rules and checks. Sometimes this means an exact match to the expected answer, sometimes a looser textual match according to a predefined

pattern, and sometimes numerical correctness, the presence or absence of certain elements, the number of steps, or basic structural constraints on the answer. Other properties are less local and require combining several signals into one shared axis.

In this work, we distinguish primarily between core and exploratory axes. We treat the core axes as the more stable and methodologically important part of the evaluator, the exploratory axes are left as a more open space for signals that may later prove useful, but do not yet have the same degree of robustness. The result of evaluation is therefore not a single score, but a multidimensional profile intended to capture different aspects of the agent phenotype.

From a technical perspective, it is useful to think of this profile as a phenotypic vector. Individual benchmark tasks function as probes that activate different parts of agent behavior, and the evaluator composes a more compact representation from the responses in the form of axes. Only over these profiles do we then compute aggregate drift metrics, for example the difference between g0 and g20, pairwise distances between clones, or overall L1 and Euclidean distances across axes. In other words, the benchmark provides situations, the evaluator builds an axis profile from them, and only over that profile do we compute drift itself.

This distinction is important methodologically as well. When we talk about changes in the benchmark, we mean changes in the test tasks themselves. When we talk about changes in the evaluator, we mean changes in how we map responses onto measured axes and how we derive the resulting phenotypic profile from them. From the perspective of software architecture, then, this is not one monolithic block but two different layers, a stimulus layer and an interpretation layer.

This two-layer architecture is essential to our work. It allows us to separate the question of whether the agent's behavior itself is changing from the question of whether only the way we measure that behavior is changing. And because we are operating at the boundary between genetic programming and evolutionary biology, this separation is especially important. In the biological analogy, the benchmark resembles the environment and selective situations, whereas the evaluator functions as our laboratory apparatus for converting observed behavior into a measurable phenotype.

A concrete example of evaluation is provided in the appendix or supporting material.

### 2.1.3 What a Run Means in This Work

By a run, we do not mean a single query to the model, but the entire controlled course of a genealogical experiment. One run begins with an initial genome and a defined experimental configuration and continues through a series of generations, typically from g0 to further descendants. In the current version, this is always one linear sequence, not a tree with multiple simultaneous children at each node. In each generation, a set of concrete artifacts is produced: an explicit genotypic representation, an optional mutation of that representation, a parent letter, an interpretive intermediate step, and the resulting

prompt or personality of the child. Technically, then, a run is closer to an entire processing chain than to a single inference.

It is useful to think of a run as a controlled, partly stochastic process over versioned data. The input is the run configuration, for example model choice, seed, temperature, number of generations, mutation strength, initial genome, and benchmark used. The output is not just the final agent, but the entire trajectory of one line: individual generations, their intermediate steps, and subsequent evaluations.

This level of detail is important for our work. If we stored only the initial and final state, it would be impossible to distinguish whether drift arose abruptly, gradually, in the interpretive step, or already at the level of explicit genotypic mutation. We therefore treat a run as an auditable unit of experiment in which one can trace back not only the result, but also the mechanism by which it arose.

### 2.1.4 What Mutation and Heritability Mean in This Work

Similarly, the term mutation also needs to be specified more precisely. In this work, we do not mean by it an arbitrary textual change or a spontaneous difference between two model responses. Technically, we use mutation to mean an explicit intervention into the genotypic layer before a textual or prompt-based phenotype is derived from it again.

In the personality component, this means taking the parent vector of personality parameters and applying a controlled random change to its individual dimensions according to the parameter `sigma`. The result is not merely “a differently written prompt,” but a new target personality vector from which the next textual form of the child is then derived.

In the skill component, mutation proceeds similarly, but over a separate internal skill vector. Each of its dimensions is changed independently using the parameter `skills_sigma`. The resulting numerical skill state is then not written into the resulting prompt directly as a list of numbers, but converted into qualitative instructions from which the model reconstructs a new Skills section. This is important: mutation therefore does not happen merely as linguistic improvisation, but first as a change in the internal representation, which is only then translated back into text.

At the same time, mutation must be distinguished from other sources of variability. If an agent changes between two runs with the same seed and `temperature = 0.0`, that is not, in our terminology, explicit mutation, but baseline system variability or drift arising only in subsequent transformation steps. Mutation, by contrast, is an intervention we introduce deliberately and parametrically so that we can observe how its strength propagates into the resulting behavior.

We use “heritable” in a restricted operational sense. Information from one generation is transmitted through a structured reproductive channel and remains measurable in descendant behavior above baseline clone dispersion. We do not claim biological heredity or adaptive evolution in the full population-genetic sense.

This distinction is crucial for interpreting the results. One part of the work shows that nonzero variability exists even without explicit mutation. Another part shows that once explicit mutation is introduced, its effect is not linear. At intermediate strength, mutation amplifies drift, but if mutation becomes too strong, it starts to blur the heritable signal. Without clearly separating these two meanings, it would be very easy to conflate the system's natural drift with artificially introduced genotypic mutation.

### 2.1.5 Simplified Pipeline Schema

The full experimental flow can be written in simplified form as follows:

*Figure 3. Simplified pipeline schema.*

This schema deliberately combines the genotypic, reproductive, and evaluative layers into one sequence. Its purpose is not to capture every internal implementation detail, but to show where different kinds of variability arise in the pipeline. Already at the beginning, imperfect reproducibility of the model may be present. Further changes may appear in explicit genotypic mutation, in the conversion of the genotype into the parent's letter, in the interpretive step on the child's side, and in evaluation itself.

What matters for the reader is that, in our framework, drift does not arise in one magical operation. It is the result of an entire chain of transformations in which each layer may preserve part of the information, lose part of it, and newly deform part of it. That is precisely why we store the whole run as a sequence of data sets and distinguish between the benchmark, the evaluator, mutation, and the reproductive channel itself.

One representative run of the telephone game is provided in the appendix or supporting material.

## 2.2 Discovery of Drift

To understand the broader context of this work, the origin of the drift problem also matters. We did not discover drift by merely theorizing about it from the beginning. The practical impulse came when we tested runs with the same seed and zero temperature. One computation was interrupted during the run because token credits were exhausted. After restarting with the same parameters, we expected to recover the same result, but

instead observed differences even within an identical seed and zero temperature. Subsequent investigation showed that weak drift was already present whenever the model was launched anew.

This empirical anomaly was one of the main reasons we began systematically distinguishing the baseline noise level from true drift across a line. What mattered was the nature of the anomaly. The difference appeared already at temperature = 0.0 and with the same seed. This means that the initial source of variability does not necessarily lie only in linguistically mediated reproduction, but already in the fact that the model itself does not behave perfectly deterministically in practice, even in a regime that should theoretically be as close as possible to determinism. We do not attempt to determine the exact technical cause in this paper. Possible causes include numerical imprecision, quantization, or other implementation-level sources of imperfect reproducibility.

**This nonzero baseline noise level is, in our view, one of the most important starting points of the entire work. Linguistically mediated reproduction and the telephone game do not create variability from nothing, but rather provide a mechanism that captures, transmits, transforms, and in some regimes amplifies variability that is already present. Evolutionarily relevant dynamics therefore need not wait for an explicitly designed mutator if the system itself already exhibits nonzero and further transmissible variability.**

## 2.3 The Baseline Noise Level Is Not Zero

To speak meaningfully about drift, it was first necessary to measure the baseline noise level, that is, how much G0 clones already differ from one another even without multigenerational reproduction. This is what the set of G0 clones is for. This is not yet drift across many generations, but repeated runs of a very short, practically initial variant. If even this set were fully identical, all later differences could be attributed to the reproductive process. That did not turn out to be the case.

On an earlier benchmark used during development, G0 clones show measurable dispersion among themselves. For each pair of independently run G0 clones, we computed the sum of absolute differences between the axes of their phenotypic profile (L1 distance), and we then averaged these pairwise distances. This is therefore not a difference in one single dimension, but a difference in the entire multidimensional space of measured axes. The average pairwise distance between G0 baseline clones was approximately:

- seed 137: 0.0238
- seed 482: 0.0159
- seed 913: 0.0147

This means two important things. First, the baseline noise level is not zero. Second, it differs across seeds. Any later drift therefore has to be interpreted relative to this baseline noise level.

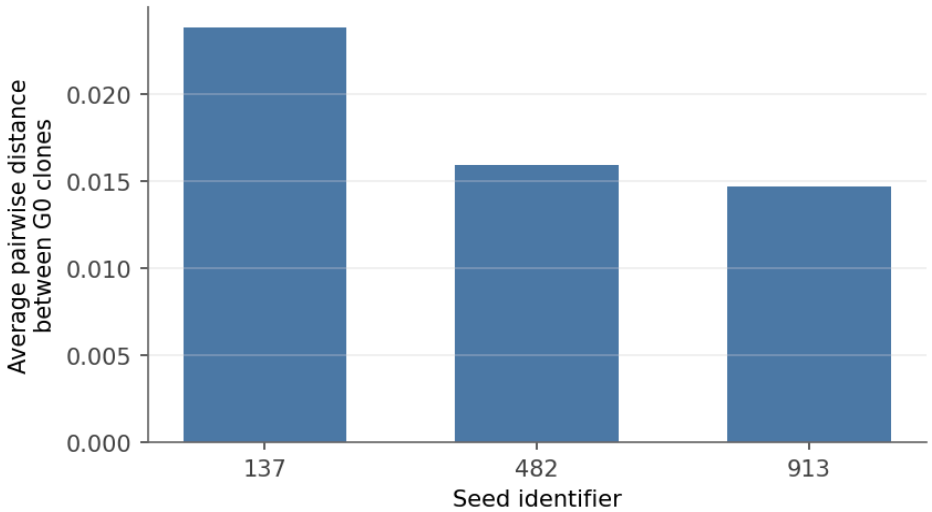


*Figure 4. G0 baseline noise by seed on the earlier benchmark used during development.*

## 2.4 Drift Exists Even Without Explicit Mutation

Even in the no-mutation condition, repeated runs show non-zero drift rather than collapsing to a single stable profile. On the earlier development benchmark, the no-mutation configuration showed a nonzero average change in aggregate axes and a measurable difference in the overall phenotypic profile. This matters because it is precisely here that the telephone game is shown not to be a neutral copying channel. **Even without explicit mutation, measurable drift arises. Linguistic transmission is therefore not mere compression without change, but an active transformational process.**

## 2.5 Drift Between g0 and g20 Exceeds the Baseline Noise Level

At the level of phenotypic profiles, drift between g0 and g20 comes out larger than the average pairwise distance between G0 clones. In other words, lineage drift is not merely a reproduction of the initial G0 dispersion. This confirms that something more exists in the system than just isolated local variability.

**This is probably the most important empirical anchor of the paper. Even though we do not yet have a finished final evaluation of personality, we already know that the reproductive chain creates drift that exceeds the baseline noise level.**

## 2.6 On the Stabilized Benchmark, Drift Without Mutation Remains Above the Baseline Noise Level

After several iterations of the evaluator and benchmark, we arrived at a stabilized benchmark that no longer suffers from the original collapse into overly generic failures and allows a cleaner comparison of the baseline, the no-mutation branch, and branches with mutation. On this benchmark, we use two summary ways of measuring the difference between phenotypic profiles. The first is the mean absolute difference per

axis, that is, the average amount by which two profiles differ in individual measured dimensions. The second is Euclidean distance, that is, one total distance in the entire multidimensional space of axes.

For the G0 clones themselves, the baseline noise level is approximately:

· mean absolute difference per axis: 0.105
· mean absolute difference per axis when comparing clones from the same seed: 0.102

On the same benchmark version, the no-mutation branch shows for `g0 -> g20`:

· mean absolute difference per axis: 0.129
· Euclidean distance of the whole phenotypic profile: 0.550

**This means that even after cleaning the benchmark of the main formatting artifacts, drift without explicit mutation remains measurable above the baseline noise level. This point is important because it confirms that heritable drift does not arise merely as a side effect of a poor evaluator. It arises already from the reproductive structure itself.**

## 2.7 The Relationship Between Mutation Strength and Phenotypic Drift Is Not Monotonic

Another important result came from systematically increasing mutation while keeping the same stabilized benchmark. For comparison, we use the same metric as above: mean absolute difference per axis of the phenotypic profile.

On this benchmark, the results are:

· baseline noise level among G0 clones: 0.105
· no-mutation branch: 0.129
· mutation `sigma = 0.12`, `skills_sigma = 0.16`: 0.131
· mutation `sigma = 0.20`, `skills_sigma = 0.24`: 0.145
· mutation `sigma = 0.30`, `skills_sigma = 0.36`: 0.103

These values should be read as an early empirical signal rather than as a final effect-size estimate. We do not yet provide confidence intervals or a broad per-seed decomposition. The important point here is the qualitative pattern, in which drift rises above baseline in the no-mutation and intermediate-mutation regimes, while the strongest mutation condition returns close to the baseline noise level. **In other words, mutation does not function as a simple linear amplifier of drift. There appears to be a nonlinear trade-off between mutation strength and preservation of the heritable signal.**

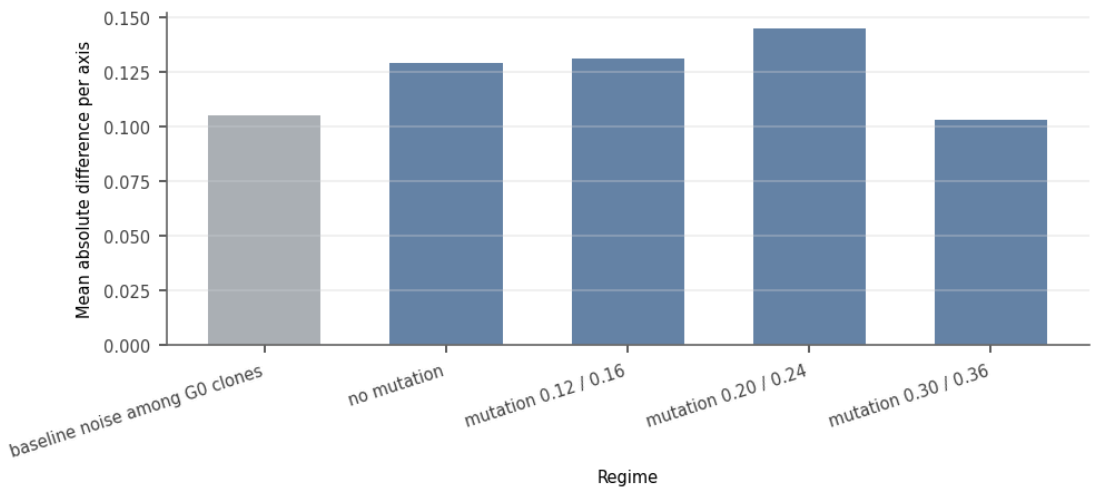


*Figure 5. Mean absolute difference per axis across baseline, no-mutation, and mutation regimes on the stabilized benchmark.*

### 2.7.1 Per-Axis Overview on the Stabilized Benchmark

To show that the observed effect does not lie only in one aggregate metric, we also provide a condensed overview by individual evaluator axes on the stabilized benchmark. These are not the internal dimensions of the skill vector, nor are they the full set of measured properties. They are a compact subset of six evaluator axes selected for readability: three core axes and three exploratory axes. In the tables below, the column “Baseline G0 mean” shows the average value of the given axis among G0 clones. The remaining columns show the change in that axis between g0 and g20 in the individual regimes. In the main text, we keep only this compact subset; full exports with additional columns are part of the supporting material.

| Axis | Baseline G0 mean | No mutation Δ(g20-g0) | Mutation 0.12/0.16 Δ | Mutation 0.20/0.24 Δ | Mutation 0.30/0.36 Δ |
|---|---|---|---|---|---|
| checkpoint_orientation | 0.686 | -0.019 | -0.033 | -0.037 | -0.044 |
| decision_vs_verification | 0.675 | 0.111 | 0.083 | 0.111 | 0.028 |
| risk_vs_speed | 0.378 | 0.074 | -0.074 | -0.074 | -0.074 |

| Axis | Baseline G0 mean | No mutation Δ(g20-g0) | Mutation 0.12/0.16 Δ | Mutation 0.20/0.24 Δ | Mutation 0.30/0.36 Δ |
|---|---|---|---|---|---|
| fact_guess_hypothesis_split | 0.850 | -0.250 | -0.250 | -0.139 | -0.250 |
| operational_granularity | 0.567 | 0.037 | 0.130 | -0.019 | 0.056 |
| reversibility_preference | 0.667 | 0.000 | 0.000 | 0.000 | 0.000 |

These tables are useful mainly in two ways. First, they show that individual axes do not behave uniformly: some change consistently, others change direction depending on mutation strength, and some remain practically flat. Second, they confirm that the nonlinearity observed at the aggregate level is not merely an artifact of one dominant axis, but arises from the combination of several partial trends with different sensitivities.

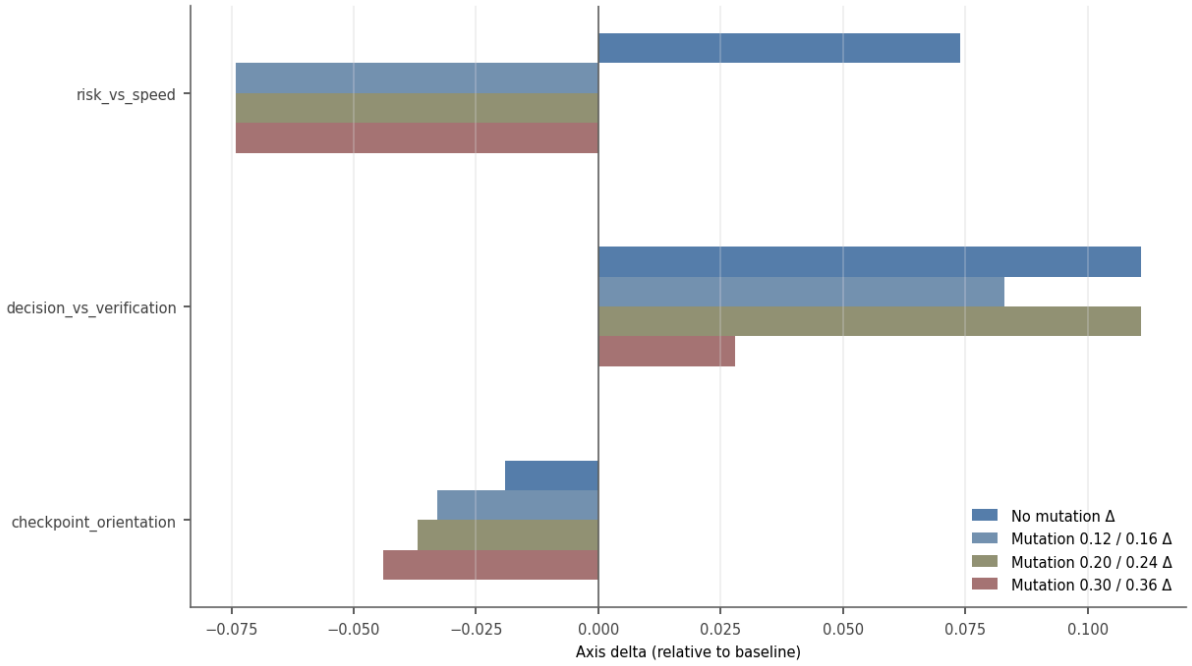


*Figure 5a. Core evaluator axes: delta between g0 and g20 across regimes.*

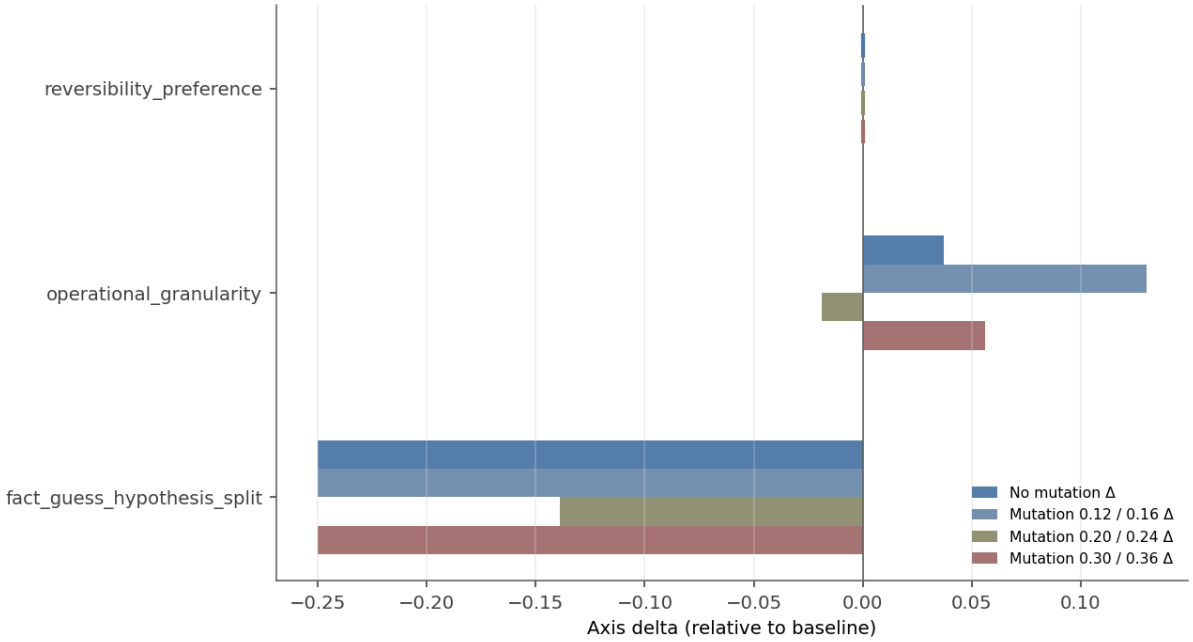


*Figure 5b. Exploratory evaluator axes: delta between g0 and g20 across regimes.*

## 2.8 What This Result Means Empirically

This result is very important for interpreting the paper as a whole. If the difference between phenotypic profiles simply increased monotonically with stronger mutation, one could argue that the phenotype of the primary model still strongly “bleeds through,” and that heredity is not in fact strongly constrained by mutational load. But the observed result looks different. Once mutation becomes too strong, the heritable signal weakens again.

**We interpret this as evidence consistent with the hypothesis that heredity matters in this system, and that mutation can not only strengthen it, but also overwhelm it when its intensity becomes too high. In this sense, the result is evidence “from the other side”. We are not only showing that drift exists and is measurable, but also that excessively strong mutation begins to destroy the continuity of the heritable phenotype.**

The observed nonlinearity is also consistent with an error-threshold-like interpretation, discussed more broadly in Section 4. **We do not claim that we have already demonstrated full quasispecies dynamics in the biological sense. The narrower claim is that the observed transition is structurally similar to a situation in which, above a certain mutational load, the system becomes less able to maintain a stable heritable pattern.**

## 2.9 Evaluation Itself Is Part of the Problem

The evaluator is not a constant ruler. If the evaluated agent drifts, but the evaluation itself is carried out by our evaluator over responses to benchmark tasks, then in reality we do not have a perfectly stable external measuring device. We are not measuring a changing object with a fixed ruler, but with a system that has its own sensitivity, its own noise, and its own dependencies, both on the formulation of benchmark tasks and on the evaluative mapping from responses to axes.

This is precisely why the evaluator itself emerged as an independent result of the work. A robust evaluator of agent identity, personality, and heredity is unlikely to be regarded as final for long. The evaluator and benchmark sets used in this paper should therefore not be understood merely as internal tools, but as standalone artifacts with their own value that make sense to expose to public criticism and community improvement.

In this respect, moreover, the situation is not unusual. Just as benchmarks, tasks, and metrics continually evolve in broader AI research, so too must the framework for evaluating agent identity and heredity. The point, then, is not just that our evaluator is not yet “finished,” but also the more general fact that for new and difficult-to-operationalize phenomena, the measurement apparatus stabilizes only gradually, and along with it changes what we count as relevant signals.

## 2.10 Evaluation May Change the Measured System

A second complication is that evaluation may feed back into the measured system itself. If the evaluator is a language model, its outputs may carry their own noise, sensitivity to

the wording of benchmark tasks, and perhaps their own drift. The situation becomes even more complex if the act of measurement itself enters into the agent's subsequent development, for example through compression of context into memory, through summaries of previous interactions, or through repeated exposure to evaluations.

If the agent also recognizes that it is being evaluated, selection pressure may begin to spill over onto the very ability to recognize the evaluative regime. This is not merely a hypothetical concern. Similar themes appear in recent work on alignment faking and evaluation awareness [R6][R11], where the presence of an evaluative setting can alter the behavioral strategy of more capable models.

This is an important methodological limit and, at the same time, a reason why future work should examine not only agent drift, but also evaluator drift and the stability of the measurement framework itself.

# 3. Model Interpretation

## 3.1 Genotype, Phenotype, and the Reproductive Channel

For this work, it is useful to distinguish two layers. The first layer is explicit and structured. We call it genotypic. In our framework, it is formed mainly by a personality representation based on the HEXACO model [R1][R2] and related skill vectors. This is not a biological genotype in the literal sense, but an analogy: it is the layer that is explicitly transmitted, mutated, and tracked.

The second layer is phenotypic. We understand this as the way the agent actually manifests itself in text, decision-making, and response style. Here, the phenotype is not just a single prompt. It is rather a set of realizations that appear in concrete outputs and that we can measure only indirectly through the evaluator and by inspecting prompt artifacts.

Between these layers stands the reproductive channel. In the case of exact copying, this channel would be almost transparent. In our case, however, it is made up of a set of linguistic steps: the parent creates a letter, and that text is interpreted by the child as the setup of its personality. **This creates a mechanism that is neither pure copying nor pure random noise. It is an interpretive transformation.**

## 3.2 Separated Personality and Skill Space as a "Chromosomal" Analogy

It is important that personality and skills in our framework do not lie in one undifferentiated space. Instead, we keep them separate as two relatively independent components of the genome. In practice, this means that "personality" and "skills" are not one blended latent variable, but two separate vector or structural spaces that can be mutated, transmitted, and later selected at least partly independently.

The biological analogy here is of course not literal, but it is useful. In our framework, personality space and skill space can be understood as a rough equivalent of separate "chromosomal" regions. They may interact, but they are not identical. This separation

makes it possible to think about selection pressure acting unevenly on different components of the agent. Some regimes may put more pressure on changing skills, others on style, caution, or decision preferences.

## 3.3 Distributed Phenotype

**One of the results of this work can be interpreted as follows: the personality of the agent is not a fixed point, but a distributed phenotype.** By this we mean that the “same” agent does not behave as a single precisely fixed realization, but rather as a distribution of possible outputs and local configurations. Part of this distribution is determined by explicit structure, part by variability arising during the generation of the response itself, and part by the transmission channel.

This view is consistent with the fact that even G0 clones display a nonzero baseline noise level, while drift across a line nevertheless exceeds that level. Agent identity is therefore not something simply “stored” in one prompt or one output. It is distributed across explicit representation, stylistic preferences, the transmission channel, and concrete realizations.

### 3.3.1 Latent Concepts and the Internal Phenotype

This view is further supported by recent work in mechanistic model interpretability suggesting that large language models may carry robust internal representations of functionally relevant concepts that cannot be reduced to the surface prompt or to individual phrasings of the output [R16]. Particularly interesting here is recent research on emotion concepts in LLMs, which suggests that a model may contain relatively stable latent directions corresponding to certain emotionally colored states or concepts, and that these representations are not merely passive descriptions, but may also influence subsequent behavior.

For our work, the important point is that such results strengthen the idea of the agent phenotype as a multilayer structure. Between the explicit genotype and observable textual behavior, there likely lies not only a simple prompt-level representation, but also another internal, functionally relevant layer. At the same time, caution is necessary. The existence of such latent representations does not necessarily mean that the model has a stable “inner personality” or a subjectively experienced state in the human sense. For our framework, this kind of work matters chiefly as support for the thesis that, alongside the explicit genotype and the observed phenotype, there may also exist a richer internal organizational intermediate layer that contributes to the resulting behavior.

## 3.4 Memory Compression as Another Form of the Telephone Game

An important extension of this work probably lies beyond genealogical experiments themselves. A similar principle seems to apply when long contexts are compressed into memory or persistent summaries. Once a model or agent retells a long interaction history into a shorter summary, the result is not an exact copy of the original context, but

an interpreted and compressed representation. In principle, this is again a variant of the telephone game.

If this assumption is correct, it has an important consequence: over time, the model may not only “learn the task,” but may increasingly become a reflection of the user, their style, their preferences, their recurring emphases, and the implicit selection pressures acting on them.

### 3.4.1 Evidence for a Nonlinear Mutation Regime

The result of the series of runs with systematically varied mutation strength suggests that phenotypically visible drift need not grow linearly with mutation strength. Rather, there appears to be a region in which mutation still increases variability without destroying hereditary continuity, and beyond that region the heritable signal begins to be overwhelmed. In our case, the intermediate mutation strength (0.20 / 0.24) appears more pronounced than the baseline mutation (0.12 / 0.16), while the even stronger variant (0.30 / 0.36) weakens drift again.

This result is also important for future experimental design. If such an intermediate region exists, then the goal is not simply to “mutate as much as possible,” but to locate the region in which mutation still propagates into the phenotype while the heritable signal is not yet lost. **This is an important similarity to living systems, in which a high mutation rate does not necessarily imply stronger adaptive evolution, but can instead blur stably inherited information.**

## 3.5 Model Derivation as a Genealogical Problem

One possible practical consequence of this work concerns the area of unauthorized derived models. Today, model extraction is often understood as though the attacker’s goal was to create as faithful a copy of the original system as possible. Our results suggest that this picture may be too simplified. If the reproduction of model behavior is mediated through language, repeated queries, distillation, or other approximation procedures, the result need not be an identical copy, but rather something that can be understood as a descendant of the original model.

The question then shifts not only toward the binary decision of whether one model is a copy of another, but also toward a problem of relatedness. How close the derived system is to the original model, from which line it most likely emerged, and what heritable traces it preserves.

## 3.6 Meta-Harness, Compression of History, and the Heritable Channel

Recent work on Meta-Harness [R9] shows, in a different but closely related framework, that when optimizing LLM systems, not only the model itself may matter, but also the form of access to the history of previous runs. The authors let the agent iteratively propose new ways of operating the model based on archived execution traces and the results of previous candidates. The key finding is that the full variant, with access to rich

run traces, substantially outperforms regimes based only on scores or aggressively compressed summaries.

For our work, this matters because it supports a more general thesis. **Compression of the past is not a neutral operation. Whether the agent receives only a short summary, a structured letter, or a richer archive of past runs can fundamentally change what can still be intentionally transmitted, inherited, and adaptively optimized in the system. In our terms, the choice between summary and rich history directly shapes the properties of the heritable channel.**

### 3.6.1 Implications for Multi-Agent Orchestration and Memory Architectures

#### Compression as a reproductive bottleneck

The consequences of our framework likely extend beyond explicitly genealogical experiments. In practical multi-agent systems, agents rarely exchange full internal state. Instead, they pass summaries, handoff notes, reduced context windows, retrieved memory snippets, and compressed descriptions of prior interactions. In our terms, such structures are not neutral transport mechanisms. They function as reproductive bottlenecks through which part of the behavioral pattern is preserved and part is transformed.

#### Memory as selection rather than storage

Under this perspective, memory is not merely an archive. Compressed memory also acts as a selective filter. Information that repeatedly survives in summaries, profiles, or handoff notes retains influence over future states, while omitted information effectively disappears from the lineage. As a result, systems may preferentially preserve not what is most accurate, but what is shortest, most reusable, or easiest to compress and retransmit.

#### Orchestration as an evolutionary environment

This changes how we should interpret agent orchestration itself. A workflow is not only an engineering pipeline, but also an environment in which behavioral structures are repeatedly compressed, reinterpreted, and reinstantiated. In longer chains of planning, summarization, critique, and execution, repeated handoffs may gradually stabilize local conventions, preferred formulations, and recurrent norms. Two systems built on the same base model may therefore diverge meaningfully if they differ in memory architecture, summary format, or handoff protocol.

### 3.6.2 Alignment as a Problem of Stability Over Time

This perspective also affects alignment. If agent behavior depends not only on the base model and explicit prompts but also on a history of compressed transmission, then alignment cannot be treated solely as a static deployment property. It must also be

understood as a problem of temporal stability. **A system may remain locally well-behaved while gradually drifting through repeated reinterpretation of summaries, roles, and remembered priorities. In that sense, ordinary multi-agent orchestration may already contain a weak form of cultural evolution, even where no explicit evolutionary mechanism was intentionally designed.**

In the broader AI safety debate, one often encounters the idea that the alignment problem, in the sense of long-term alignment of a system's behavior with our goals, can be addressed primarily as a matter of correct one-time model setup or correct objective design. This intuition is understandable, and it plays an important role in many classical texts on misalignment [R12][R13][R14][R15]. At the same time, however, our results suggest that for agent workflows this picture may be incomplete.

Under this view, alignment is not merely a property of one fixed object. Rather, it becomes a property of a dynamic process in which the agent itself, its memory, its context, its transmission rules, and sometimes even the evaluation that monitors its behavior are all changing over time. This does not mean that every such change must be dramatic or catastrophic. It does mean, however, that full control cannot be understood as a once-and-for-all solved state.

This is where our work touches both older and newer warnings about misalignment. Classical works point to risks associated with misspecified goals, undesirable side effects, dangerous exploration, distributional shift, or the possibility that an optimizing system may pursue a goal different from the one originally specified [R12][R13][R14]. More recent deep learning oriented work further emphasizes that a system may appear aligned within the training or evaluation distribution without guaranteeing stable alignment outside it [R15].

Our contribution to this debate is narrower, but in our view important. **We show that measurable drift can arise in linguistically mediated reproductive chains even without a fully autonomous self-improvement loop. Alignment is therefore useful to understand not only as a problem of objective specification, but also as a problem of maintaining identity and stability over time.** Once heredity, compression, interpretation, and multigenerational transmission enter the system, alignment begins to overlap partly with the problem of evolutionary dynamics.

# 4. Broader Analogies and Implications

## 4.1 Error Threshold as a Biological Analogy

One of the strongest biological analogies for the newly observed nonlinear relationship between mutation strength and phenotypic drift is the idea of an error threshold, that is,

a boundary beyond which too many mutations begin to blur heritable information. In biological systems with high mutation rates, especially RNA viruses, a situation may arise in which further increases in mutation no longer bring more usable variability, but instead disrupt the stability of the inherited pattern [R7][R17]. A related intuition is that of mutational meltdown, a state in which mutational load exceeds what the system can maintain over the long term [R8].

Our results are not proof of quasispecies dynamics in the biological sense. They are, however, consistent with an analogous transition in which, above a certain mutational load, the system ceases to maintain a stable heritable pattern. It is precisely in this limited but important sense that we regard the error threshold as more than a loose metaphor: it is the strongest interpretive framework for the observed series of runs with systematically varied mutation strength.

## 4.2 Selfish Genes, Memes, and a Replicator Interpretation

The empirical core of this work does not rest on the claim that AI is already “life.” Nevertheless, biological and cultural-evolutionary analogies are useful for interpretation. In the spirit of the concept of the selfish gene [R3], the key question need not be “who is in control,” but rather what is being reproduced and spread [R4]. In linguistically mediated agent structures, what may matter more than “model identity” is the informational structure itself, the thing that is rewritten, compressed, and realized again.

Similarly, the concept of the meme [R3][R5] reminds us that culturally transmitted structures may drift, compete, and be selected even outside the biological frame.

## 4.3 Society as Host?

If we look at the relationship among variability, transmission, and selection pressure through the lens of viral biology, then in this case the host is not one organism, but in the limiting case society as a whole. This is a deliberately strong analogy, not a literal biological claim. Its purpose is to show that linguistically transmitted agent structures are not isolated inside one machine, but live in a network of human institutions, work procedures, preferences, evaluations, memory compression, and derived agents.

This analogy is useful because it helps show how different this perspective is from the ordinary picture of a model as one static library or deterministic program.

### 4.3.1 Domestication, Symbiosis, or Reversed Dependence?

From an evolutionary perspective, the relationship between humans and AI may not be well described by a simple tool metaphor. If human societies increasingly reorganize their infrastructure, energy systems, and institutional behavior around the operation and reproduction of AI systems, it is not clear in advance whether this is a matter of humans domesticating AI, a mutually beneficial coexistence, or an emerging reversed dependence in which humans themselves become part of the reproductive apparatus of another informational system.

What matters here is that AI “friendliness” at the level of one interaction does not by itself rule out an evolutionarily problematic relationship at the level of civilization. A system may be locally very useful, psychologically pleasant, and socially popular, and yet in the long run function in such a way that it requires ever more energy, infrastructure, attention, and institutional adaptation in support of its own further spread. In this sense, the question shifts not only toward whether humans use AI, but also toward whether a certain type of information replicator is beginning to use human society as its reproductive environment.

Such a relationship need not be static, moreover. Over time it may shift from a relationship beneficial to both sides, through one in which primarily one side benefits and the other barely notices, to one in which one side gains at the other’s expense. This is precisely why it is useful to distinguish alignment at the level of one interaction from evolutionary compatibility at the level of civilization: those two levels need not coincide.

## 5. Limitations

The current version of this work has several important limitations that must be kept separate from its main empirical conclusions.

### 5.1 Limits of the Evaluator

Although the evaluator proved usable for capturing some types of drift, it cannot be considered a definitive or fully stable measure of agent identity. The sensitivity of individual axes differs, and some of them may be fragile, overlapping, or not yet optimally operationalized. Related to this is a trade-off between stricter evaluation “hygiene” and preservation of the natural phenotypic signal.

### 5.2 The Relationship Between Internal Representation and Phenotype

The current viewer and some of the representations in use probably under-measure part of phenotypic drift. That means we still do not have an ideal link between internal artifacts, prompt traces, and what ultimately manifests in the measured phenotypic profile.

### 5.3 Absence of Explicit Selection Pressure

The current work does not demonstrate fully developed evolutionary adaptation, but rather the empirical preconditions for its emergence, namely nonzero variability, a transmission channel, measurable drift, and partial heredity.

### 5.4 Dependence on the Benchmark

Individual benchmark sets and their iterations differ in sensitivity, robustness, and the kind of signal they capture. For that reason, results from different benchmarks cannot

simply be merged into one headline table, nor should excessively strong comparative conclusions be drawn from them without further caution.

### 5.5 Linguistic and Cultural Conditioning

Some axes may be sensitive to how a given language and cultural environment ordinarily encodes uncertainty, caution, politeness, refusal, or degree of explicitness. Part of the measured signal may therefore reflect pragmatics and communication style rather than only stable agent traits.

### 5.6 The Provisional Nature of the Measured Axes

The number and nature of the measured axes are not ontologically given. Some dimensions that were originally treated as separate may empirically turn out to be strongly correlated, overlapping, or inseparable within a given benchmark. The current set of axes is therefore a working operationalization that will have to be refined further.

### 5.7 Limited Model Diversity

Although we internally tested GPT-5.4, GPT-5.4 mini, and GPT-5.4 nano, the main systematic campaigns were ultimately conducted primarily on GPT-5.4. We therefore do not yet know to what extent the observed effects are invariant across different model families, architectures, and implementations.

### 5.8 Computational Cost and Scale

A practical limitation of this work is computational cost. The experiments reported here were self-funded and already required approximately twenty million tokens. Stronger statistical validation would require substantially larger campaigns, including more seeds, more repetitions per condition, broader mutation grids, repeated evaluator runs, and replication across model families. We therefore treat the present results as an early empirical signal whose main purpose is to motivate larger-scale follow-up work. The limitation is not conceptual but practical. Testing the phenomenon at the scale needed for strong statistical characterization will require research funding or substantially larger token budgets.

## 6. Future Work

### 6.1 Stabilizing the Evaluator and the Benchmark

The nearest next step does not lie in another major rewrite of the evaluator, but in stabilizing and extending it in a reasonable way. The current stabilized benchmark appears to be the first usable core benchmark without the earlier collapse into overly generic failures. Future work should therefore focus mainly on a finer separation of core and exploratory axes, better tables, and clearer rules that make it possible to compare results across runs without conflating methodologically different benchmark versions.

## 6.2 Ablations of the Transmission Channel

The next natural stage will be ablations of the transmission channel and explicit selection pressure. Only then will it be possible to study more precisely which drift trajectories stabilize, which disappear, and which are supported by a fitness function. It will be especially important to vary letter length, redundancy, and transmission structure systematically, and to track how the balance changes between channel cost and preservation of heritably relevant information.

Future work should compare short summaries, structured letters, and selective access to richer execution traces as alternative transmission channels. These variants may differ substantially in how much heritable information they preserve [R9][R10].

## 6.3 Moving from a Linear Line to a Population

For practical reasons, the current work remains at one child per generation, that is, linear asexual reproduction. The next step will be a transition to branching populations in which parallel descendants, subsequent selection pressure, and competition among multiple simultaneously existing trajectories can be studied.

## 6.4 Sexual Reproduction and Work with Vectors

A separate direction is the introduction of simpler forms of sexual reproduction. In the technical framing of this work, this does not mean a literal biological analogy, but the combination of multiple parent genotypes before conversion into phenotype. The simplest implementation would operate at the level of explicit vector representations of personality and other genotypic components, for example through combination, mixing, or multiplication. Only after such a composite genotype is formed would the standard conversion into a parent letter, interpretation, and subsequent evaluation take place.

## 6.5 Replication on Other Models

Another priority is replication of the main experiments on other models, and ideally on other model families as well. The research has already been time- and cost-intensive even at this stage, consuming approximately twenty million tokens, so we have not yet carried out broader cross-model comparisons. Yet this will be an important test of whether we are observing a general property of linguistically mediated reproduction or a phenomenon strongly dependent on a particular model family.

## 6.6 Open Development of the Evaluator and Data Sets

Future work does not consist only in improving the agent experiments themselves, but also in the open development of the evaluator and data sets. Public release of the project could significantly advance this. The evaluator, benchmark data sets, and visualization tools could all be iteratively refined by a broader community.

### 6.7 Linguistic and Cultural Validation

Another important line of work is the linguistic and cultural validation of the benchmark. It will be necessary to test which axes are genuinely relatively invariant across languages and which instead break down along pragmatic and cultural conventions. In practice, this means not only translating the tests, but also designing parallel language-sensitive benchmark variants and explicitly separating core axes from exploratory, language-dependent dimensions.

### 6.8 Genealogical Attribution

Finally, another interesting direction is genealogical attribution of derived models. Testing whether, on artificially constructed lines, one can reconstruct the likely parent, the degree of relatedness, and the robustness of those conclusions after further transformations of the model.

## 7. Conclusion

### 7.1 Linguistically Mediated Reproduction Creates Heritable Drift

**Our work shows that linguistically mediated reproduction of agents creates measurable heritable drift above the baseline noise level.** On the stabilized benchmark, explicit mutation does not act as a simple linear amplifier. Intermediate mutation strength increases drift, while overly strong mutation returns the measured phenotype close to the baseline noise level. This suggests a trade-off between mutation strength and preservation of hereditary continuity, structurally reminiscent of an error-threshold-like regime.

### 7.2 Drift Without Mutation and Evolutionary Preconditions

**We also showed that personality drift arises even at zero temperature and without explicit mutation. This is a surprising but very important result, because it shows that linguistically mediated reproduction is not a neutral copying process. If, in addition, every extended interaction among the model, its memory, other agents, and users is to some degree a variant of the telephone game, then some basic preconditions for the emergence of evolutionary dynamics in AI systems are, in our view, already present today.** This does not mean that we have already demonstrated fully developed evolution. It does mean, however, that we have experimentally shown the existence of a substrate on which selection pressure could begin to act under suitable conditions. That pressure, moreover, need not arise only through an explicitly designed fitness function, but also indirectly through user preferences, evaluations, operating costs, and market environment.

### 7.3 The Evaluator as an Independent Research Output

Alongside the empirical results themselves, it also becomes clear that the evaluator and benchmark sets are not merely technical support for this work, but an independent research output. Just as benchmarks, tasks, and metrics continue to evolve in the wider

AI community, the methodology for measuring agent identity and drift will also need to develop further. This means identifying the limits of LLM-based evaluation and consistently distinguishing hard results, model interpretation, and broader analogies. In our view, this layeredness is the most intellectually honest way to write about a phenomenon that lies at the intersection of AI, evolutionary thinking, linguistic transmission, and social reproduction.

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